

Self-improving Cost-aware Retrieval-Augmented Generation (RAG): A Compute Allocation Perspective

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Abstract

This report conceptualises a research direction for self-improving, cost-aware retrieval-augmented generation (RAG) systems. The central idea is to model a RAG pipeline as a budgeted sequential decision problem with multiple compute knobs (reformulation, retrieval, reranking, fusion, reasoning, verification). We propose: (i) component-level scaling plots per workload; (ii) a learned controller that allocates compute under constraints; (iii) instrumentation and replay/OPE for safe self-improvement; (iv) an evaluation suite and experimental matrix; and (v) a prototype architecture (“RAGtune”) as a RAG control plane.

1 Executive summary

RAG combines parametric generation with non-parametric retrieval to improve knowledge-intensive tasks and provide provenance [20]. Industry RAG faces hard constraints: token budgets, latency SLOs, governance, and explicit per-token pricing [25, 2, 10]. This motivates treating RAG as compute allocation: choose which component to scale for the next marginal gain (vs always scaling the reasoner).

2 Motivation and industrial relevance

Scaling laws show predictable improvements with training compute/model size [16, 12], but retrieval augmentation can substitute for parameter scaling in some regimes [4]. Cloud vendors ship managed RAG building blocks and explicitly document token, latency, and governance challenges [3, 11, 25].

3 Literature and ecosystem survey

3.1 Core RAG, retrieval, fusion, reranking

Canonical methods: RAG [20], DPR [17], FiD [13], MonoT5 reranking [28], late interaction (ColBERT) [18], and heterogeneous retrieval benchmarking (BEIR) [32].

3.2 Reformulation and fusion

HyDE reformulates via hypothetical document generation [9]. Multi-run fusion can use Reciprocal Rank Fusion [6].

3.3 Adaptive retrieval, robustness, evaluation

Adaptive retrieval and gating: Self-RAG [1], RetrievalQA [38], and uncertainty-to-retrieval analysis [27]. Robustness to irrelevant context [37]. Retrieval evaluation aligned to downstream performance: eRAG [31].

3.4 Cost-aware inference

FrugalGPT illustrates cost-aware cascades/routing over LLM APIs [5] and motivates analogous controllers over pipeline knobs.

3.5 Industrial products

Managed RAG: Bedrock Knowledge Bases [3], Vertex AI RAG Engine [11], Azure AI Search RAG/agent retrieval [25]. Ecosystem tools: LangChain [19], LlamaIndex [22], vector DB docs [35, 30].

4 Component-level scaling plots

Define knobs $\mathbf{k} = (k_{ref}, k_{ret}, k_{rer}, k_{fus}, k_{rea}, k_{ver})$ and performance/cost:

$$\text{Perf} = f(\mathbf{k}; w), \quad \text{Cost} = g(\mathbf{k}; w). \quad (1)$$

FiD provides linear scaling in #passages fused [13]. Reranking cost scales with #query-doc pairs scored (MonoT5) [28]. Reasoning can scale at test time via sampling/search (self-consistency, ToT) [34, 36].

4.1 Example plot (illustrative)

Figure 1 is a placeholder sketch; empirical curves are to be measured per workload.

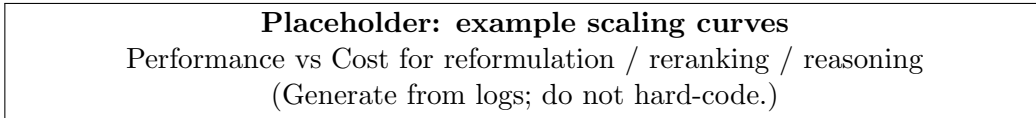


Figure 1: Example scaling curves (placeholder).

5 Estimating curves per workload

Use structured knob sweeps and proxy evaluation (eRAG) to reduce evaluation cost [31]. Account for retrieval infrastructure (ANN): FAISS [15] and HNSW [23]. Measure robustness because retrieval can hurt [37].

6 Scheduler/controller formalisation

Let the controller choose actions a_t given state s_t under budgets. Learn policies with contextual bandits and offline evaluation. Theoretical foundations: unbiased LTR from biased feedback [14], doubly robust OPE [8, 7], contextual-bandit online LTR [21].

6.1 Mermaid flowchart (source)

```
graph TD
    A[Start: new query] --> B{Risk tier / compliance?}
    B -->|High| C[Enable citations + verifier budget]
    B -->|Normal| D[Default budgets]
    C --> E[Initial retrieval (small k)]
    D --> E
    E --> F{Evidence quality OK?}
    F -->|No| G[Reformulate / decompose]
    G --> H[Retrieve again or increase k]
    H --> I[Optional rerank deeper]
    F -->|Yes| I
    I --> J{Need multi-hop / synthesis?}
    J -->|Yes| K[Increase fusion passages or reasoning compute]
    J -->|No| L[Use SLM / low compute]
    K --> M[Generate answer]
    L --> M
    M --> N{Verifier passes?}
    N -->|Fail| O[Escalate: more evidence or bigger reasoner]
    N -->|Pass| P[Return final + citations]
    O --> E
    P --> E
```

7 Instrumentation and self-improvement loop

Log full trajectories for replay and OPE; store propensities for exploration. Use proxy signals: eRAG doc utility [31], factuality via FActScore [26], claim verification via FEVER [33], sampling-based hallucination detection [24].

8 Evaluation suite and experiments

Benchmarks: KILT [29], BEIR [32], RetrievalQA [38], robustness to irrelevant context [37]. Report Pareto frontiers and cost-to-target metrics (quality at budget, budget for target quality).

9 RAGtune prototype design and roadmap

RAGtune is a RAG control plane: component registry, policy runtime, observability store, offline optimiser, Pareto compiler, and dashboards. Integrate with managed backends: Bedrock KB [3], Vertex RAG Engine [11], Azure AI Search [25].

References

- [1] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024.
- [2] AWS. Amazon bedrock pricing, 2026. Accessed 2026-03-05.
- [3] AWS Documentation. Knowledge bases for amazon bedrock (rag), 2026. Accessed 2026-03-05.

- [4] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, et al. Improving language models by retrieving from trillions of tokens. In *International Conference on Machine Learning*, 2022.
- [5] Lingjiao Chen, Matei Zaharia, and James Zou. Frugalgpt: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.
- [6] Gordon V. Cormack, C. L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of ACM SIGIR*, 2009.
- [7] Miroslav Dudík et al. Doubly robust policy evaluation and optimization. *Statistical Science*, 2014.
- [8] Miroslav Dudík, John Langford, and Lihong Li. Doubly robust policy evaluation and learning. In *ICML*, 2011.
- [9] Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. Precise zero-shot dense retrieval without relevance labels. *arXiv preprint arXiv:2212.10496*, 2022.
- [10] Google Cloud. Vertex ai generative ai pricing, 2026. Accessed 2026-03-05.
- [11] Google Cloud Documentation. Vertex ai rag engine overview, 2026. Accessed 2026-03-05.
- [12] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, et al. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, 2022.
- [13] Gautier Izacard and Edouard Grave. Leveraging passage retrieval with generative models for open domain question answering. *arXiv preprint arXiv:2007.01282*, 2021.
- [14] Thorsten Joachims, Adith Swaminathan, and Tobias Schnabel. Unbiased learning-to-rank with biased feedback. In *Proceedings of ACM WSDM*, 2017.
- [15] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with gpus. *arXiv preprint arXiv:1702.08734*, 2017.
- [16] Jared Kaplan, Sam McCandlish, Tom Henighan, et al. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- [17] Vladimir Karpukhin, Barlas Oguz, Sewon Min, et al. Dense passage retrieval for open-domain question answering. In *Proceedings of EMNLP*, 2020.
- [18] Omar Khattab and Matei Zaharia. Colbert: Efficient and effective passage search via contextualized late interaction over bert. *arXiv preprint arXiv:2004.12832*, 2020.
- [19] LangChain. Langchain retrieval: Build rag applications, 2026. Accessed 2026-03-05.
- [20] Patrick Lewis et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, 2020. NeurIPS 2020.
- [21] Cheng Li et al. A contextual-bandit approach to online learning to rank. *arXiv preprint*, 2019.
- [22] LlamaIndex. Introduction to rag (llamaindex documentation), 2026. Accessed 2026-03-05.
- [23] Yu. A. Malkov and D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *arXiv preprint arXiv:1603.09320*, 2016.

- [24] Potsawee Manakul, Adian Liusie, and Mark Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. *arXiv preprint arXiv:2303.08896*, 2023.
- [25] Microsoft Learn. Retrieval-augmented generation (rag) in azure ai search, 2026. Last updated 2026-01-15; accessed 2026-03-05.
- [26] Sewon Min, Kalpesh Krishna, Xinxu Lyu, et al. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation. In *EMNLP*, 2023.
- [27] Shuo Ni et al. When do llms need retrieval augmentation? mitigating overconfidence in llms. In *Findings of ACL*, 2024.
- [28] Rodrigo Nogueira, Zhiying Jiang, Ronak Pradeep, and Jimmy Lin. Document ranking with a pretrained sequence-to-sequence model. In *Findings of EMNLP*, 2020.
- [29] Fabio Petroni, Aleksandra Piktus, Angela Fan, et al. Kilt: a benchmark for knowledge intensive language tasks. In *NAACL-HLT*, 2021.
- [30] Pinecone. Build a rag chatbot (pinecone docs), 2026. Accessed 2026-03-05.
- [31] Alireza Salemi and Hamed Zamani. Evaluating retrieval quality in retrieval-augmented generation. *arXiv preprint arXiv:2404.13781*, 2024.
- [32] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. *arXiv preprint arXiv:2104.08663*, 2021.
- [33] James Thorne et al. Fever: a large-scale dataset for fact extraction and verification. In *NAACL-HLT*, 2018.
- [34] Xuezhi Wang, Jason Wei, Dale Schuurmans, et al. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.
- [35] Weaviate. Retrieval augmented generation (rag) (weaviate docs), 2026. Accessed 2026-03-05.
- [36] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. *arXiv preprint arXiv:2305.10601*, 2023.
- [37] Ori Yoran et al. Making retrieval-augmented language models robust to irrelevant context. *arXiv preprint arXiv:2310.01558*, 2023.
- [38] Zihan Zhang, Meng Fang, and Ling Chen. Retrievalqa: Assessing adaptive retrieval-augmented generation for short-form open-domain question answering. In *Findings of ACL*, 2024.